



Amrita School of AI
COIMBATORE CAMPUS

SESSIONS 1-3

What the cloud actually is

Strip away the marketing and one idea is left.

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Someone else owns the hardware, and you rent it by the minute through an API.

Everything else follows from that sentence.

Buying a server vs calling an API

● Before

- Forecast peak demand
- Raise capital, get approval
- Order, wait six weeks
- Rack it, cool it, power it
- Depreciate over five years
- Own the idle time

● After

- `POST /instances`
- It exists in 30 seconds
- `DELETE /instances`
- You stop paying
- Someone else owns the idle time

Five characteristics. All five, or it is not cloud.

1. On-demand self-service
2. Broad network access
3. Resource pooling
4. Rapid elasticity
5. Measured service

A dedicated server rented for a year fails **elasticity** (capacity is fixed) and **measured service** (flat fee). That is **hosting**. Useful, but a different thing.

Who is responsible for what

	on-prem	IaaS	PaaS	SaaS
apps	you	you	you	them
data	you	you	you	them
runtime	you	you	them	them
OS	you	you	them	them
virt	you	them	them	them
svrs	you	them	them	them
dc	you	them	them	them

More abstraction upward.

Less control upward.

Neither end is “better”. You are choosing what you want to stop thinking about.

TWO MORE, NOW MORE COMMON THAN CLASSIC PAAS

CaaS and FaaS

CaaS — containers as a service

You supply an image. The platform runs and scales it.

Cloud Run, ECS, AKS

FaaS — functions, “serverless”

You supply a function. Billed per invocation.

Lambda, Azure Functions

SERVERLESS MEANS NO SERVERS YOU THINK ABOUT

Where the machines are

MODEL	FITS
public	variable demand
private	regulated data, steady load
hybrid	steady base + bursts
community	shared sector concerns

The university runs **hybrid** without calling it that:
teaching and research on our own cluster, plus rented GPUs when a deadline needs sixty of them for a week.

THE PART NOBODY PUTS ON A SLIDE

DO THE ARITHMETIC

A 4x A100 node

Own it ~\$100 000, four-year life = **\$2.90 / hour** if it runs continuously

Rent it = **\$12-15 / hour** on demand

Break-even near **25% utilisation**.

Above that, owning wins on hardware alone.

Then add power, cooling, space, and the person who administers it, which institutional comparisons usually forget.

When each one wins

● Cloud wins

- Bursty or unpredictable demand
- Short projects
- Hardware you cannot justify buying
- You need it **today**
- Global reach without a building

● Owning wins

- Steady, high utilisation
- Predictable multi-year load
- Data that cannot leave
- Sustained egress-heavy work

Egress

Data **in** is free.

Data **out** is \$0.08-0.12 per GB.

10 TB of results $\approx$ **\$1 000** to retrieve.

This asymmetry is deliberate. It is the primary mechanism of lock-in, and it is more effective than any proprietary API.

Region, zone, edge

Region — a geographic area (`ap-south-1`, Mumbai)

Availability zone — isolated datacentre within a region

Edge — small point of presence for delivery

Spread across **zones** for availability: they fail independently and are cheap to span.

Spread across **regions** for disaster recovery and data residency: expensive, and latency is physics.

Three kinds, and one common mistake

KIND	LATENCY	FOR
object	~100 ms	datasets, backups
block	<1 ms	OS disks, databases
file	~1 ms	shared directories

Object storage is a ****key-value store over HTTP****, not a filesystem. Flat namespace. Immutable objects. "Folders" are a prefix convention.

Your own network, inside theirs

A **VPC** gives you address ranges, subnets, routing, gateways.

Public subnets reach the internet. Private ones reach out through NAT.

	SECURITY GROUP	NETWORK ACL
attaches to	instance	subnet
state	stateful	stateless
rules	allow only	allow + deny

The provider secures the cloud. You secure what you put in it.

Nearly every reported “cloud breach” is a customer misconfiguration.

What actually prevents incidents

- **Least privilege**, no shared root
- **MFA** everywhere, no exceptions on admin
- **Encrypt** at rest and in transit
- **Never commit secrets** — public keys are scanned and abused in minutes
- **Log and audit** — you cannot investigate what you did not record
- **Private by default**

Next: how one machine becomes many

Virtualization.